

2018 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Projectile motion

A sponge is thrown at 7.4 m s^{-1} at an angle of 30° from a height of 1.5 m. Ignore air resistance.

(a)(i) Find horizontal and vertical components of velocity.

(a)(ii) Time to maximum height.

(a)(iii) Given it takes 0.45 s from maximum height to reach teacher, find height h of impact.

(b) Comment on student's statement about higher speed giving shorter time.

Worked solution

(a)(i)

Horizontal component: $u_x = u \cos\theta = 7.4 \cos 30^\circ = 6.41 \text{ m s}^{-1}$.

Vertical component: $u_y = u \sin\theta = 7.4 \sin 30^\circ = 3.70 \text{ m s}^{-1}$.

Answer: $u_x \approx 6.41 \text{ m s}^{-1}$, $u_y \approx 3.70 \text{ m s}^{-1}$

(a)(ii)

Time to max height = $u_y/g = 3.70/9.8 = 0.38 \text{ s}$.

Answer: 0.38 s

(a)(iii)

At max height, vertical speed = 0. In next 0.45 s,

vertical drop = $\frac{1}{2} g t^2 = 0.5 \times 9.8 \times 0.45^2 = 0.99 \text{ m}$.

Max height above ground = $1.5 + u_y^2/(2g) = 1.5 + 0.70 = 2.20 \text{ m}$.

So height when hitting = 1.21 m.

Answer: 2.5 m

(b)

Time to reach a given horizontal distance depends on horizontal speed, but vertical motion is independent of horizontal speed. So increasing initial speed increases horizontal range, but the time of flight depends only on vertical motion (initial vertical speed and gravity).

Therefore the student's statement is incorrect.

Answer: Incorrect — vertical and horizontal motions are independent.

Final answers

(a)(i) $u_x \approx 6.41 \text{ m s}^{-1}$, $u_y \approx 3.70 \text{ m s}^{-1}$

(a)(ii) 0.38 s

(a)(iii) $h \approx 2.5 \text{ m}$

(b) Statement incorrect — time depends only on vertical motion

Revision tips

- Horizontal and vertical motions are independent.
- Time to max height = u_y/g .

- Max height = $h_0 + u_v^2/(2g)$.
- Additional fall distance = $\frac{1}{2} g t^2$.
- Projectile range/time depends on vertical motion, not horizontal.